Ubiquity Symposium

Artificial Intelligence

Foundational Technologies of Artificial Intelligence

by Jeffrey Thompson, Matthew Yaffe, Odlyzko Merigo, Kemal Phil Picton

Editor's Introduction

For over seventy years since its inception in the early 1950s, artificial intelligence (AI) has advanced to perform cognitive tasks that were once believed to be exclusive to humans. This development is rooted in a diverse foundation of technological innovations from computer science and theoretical concepts in fields such as philosophy, psychology, logic, game theory, and cognitive science. Key enabling technologies for AI encompass search algorithms, logical reasoning, neural networks, natural language processing, signal processing, computer graphics, programming, software engineering, human-computer interaction, communications, and specialized supercomputing hardware. The concept of Artificial General Intelligence, defined as intelligence matching or surpassing the human brain, remains largely aspirational and is unlikely to be realized in the near future. Notably, the various technologies that constitute AI have been designed to solve specific problems, rather than directly replicate human intelligence..

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Over the past seventy years since AI emerged as an academic discipline in the early 1950s, it has led to the development of numerous technologies capable of performing cognitive tasks once thought to be solely the domain of humans. This evolution was shaped by a broad foundation of innovations in computer science as well as intellectual influences from philosophy, psychology, logic, game theory, and cognitive science. In this essay, we aim to outline the foundational technologies that enable the advancement of artificial intelligence. They include:

- Search; looking for solutions to problems in a space of possible answers
- Reasoning; applying rules and facts to make logical deductions
- Neural networks; systems of artificial neurons that learn from data
- Natural language processing; systems that analyze text and discern its meanings

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- Signal processing and computer graphics
- Programming and software engineering; almost all AI systems use conventional software engineering to assemble reasoning and machine learning modules into useful systems.
- Human-computer interaction (HCI); systems designed to interact with people
- Communications; networked communication including mobile phones and the internet
- Hardware; highly parallel chips that bring supercomputing power to the technologies referred to above

Each of these domains is rich in its own complexities. We have, however, found a general awareness of these technologies is usually sufficient to understand how AI services and products work—and therefore to evaluate their weaknesses and strengths.

A Sketch of the Foundational Technologies

Search. You have mislaid your keys. One way to search for them is to identify every possible place they could be (call this the "search space") and examine each and every location. This is called the "brute force" approach—it would eventually find your keys but would be impractically slow for a large search space. A better way would be to look in all the places you remember being recently—this heuristic, or rule of thumb, approach can reduce the search time significantly, but it will sometimes fail. In his pioneering 1950 paper on chess, Claude Shannon advocated a heuristic that combined "depth-first search" (looking far ahead at a few possibilities) with "breath-first search" (looking a few moves ahead for all possibilities) [1]. For many problems the space of all possible solutions is enormous and heuristic computer searches are the only feasible approaches.

An example is the Page Rank algorithm developed in the 1980s by Larry Page and Sergey Brin for their fledgling Google internet search engine. Rather than search for every document containing the keywords, they narrowed the search to frequently cited documents. This heuristic drastically reduced the search time and, when commercialized, made a fortune for its inventors. Many AI problems involve huge search spaces that are beyond the reach of human intellects. Heuristic searches can explore where no human has gone before.[1]

One widely used heuristic for search involves choosing a possible solution at random and inspecting similar solutions to see if they are better. If so, accept the best and inspect similar solutions to them. And so on until the solution cannot be improved. This is called "hill climbing" by analogy with a blindfolded person climbing a hill. By testing the ground with their toes, the climber can find better solutions that are farther up the hill. This will get them to the top of the local hill they are on. But many search spaces are similar to a mountain range, and there may be

[1] By this we mean that machines can enable humans to explore huge spaces of possibilities. This practical amplification of human intelligence has been a reality for many years. Whether or not machines can initiate searches to amplify their own intelligence is an unresolved question.

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a higher hill. Another heuristic called simulated annealing can help here. This involves making random jumps to more distant solutions, which are likely to place the climber on a better hill. The problem with this is that it is also possible to jump to a worse hill. The heuristic of taking big jumps at first followed by progressively smaller jumps increases the likelihood of a good solution but does not guarantee it. Genetic algorithms provide another heuristic for exploring large search spaces. This involves representing possible solutions as strings of numbers analogous to chromosomes. The search process begins by creating a population of chromosomes and successively combining parts of chromosomes with parts of others. The best (or 'fittest') solutions are retained for further iterations of the combination process. This approach corresponds to the combination of genes in biological reproduction. In principle the best parts of two mediocre solutions may be combined to create a much better solution. To ensure that possibilities outside the original random population are included, a process analogous to mutation is introduced that randomly changes some of the numbers.

Reasoning. Early AI research used logic and symbolic computation to program systems that appeared to use logic to solve problems. This was called "elementary reasoning" because it mimicked the way humans explain and justify solutions to problems. An early example was Terry Winograd's SHRDLU program that let users move blocks on a playing field by issuing natural language commands. This system had to distinguish terms such as "behind" and "on top of" and be able to plan paths so that a moving block did not crash into another. Generalizing this to more complex real-world problems require detailed, accurate models of the targeted world. But getting machines to build models of messy complex real-world situations is an unresolved scientific challenge and currently a major limitation of AI. In contrast, considerable progress has been made automating reasoning and logic for systems that can prove mathematical theorems or make expert medical diagnoses. These systems work well because their rules are well defined and easily captured into working systems and can be applied without requiring the machine to understand the meanings of any symbols or terms. Early reasoning programs used specialist languages, such as Lisp and Prolog, to represent functions and logical deductions. Today a wide variety of programming languages is used in machine reasoning and symbolic AI including Python, R, Java, JavaScript, C++, and Julia.

In 1982 the Japanese launched their multimillion-dollar Fifth Generation project. They intended to build supercomputers that could execute large Prolog programs and expand the space of problems to which reasoning could be applied. The U.S. responded with its "strategic computing initiative," which emphasized highly parallel supercomputers solving a wide range of difficult scientific problems, not just logic and reasoning problems. Although Japan did not succeed in boosting their position with reasoning machines, they stimulated an international movement to develop supercomputing approaches throughout science. Supercomputing is essential to the development of AI.

Neural networks. Neural networks are computational structures that model the way neurons in the brain fire and interconnect. Initially their performance was too slow for neural networks to

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become part of mainstream computing. But they had an advantage that logic-based computing could not match. Neural networks could learn an input-output function by being shown examples. In 1970, Seppo Linnainmaa invented a training algorithm called "back propagation" that over the next decade was perfected into a highly efficient method for training large networks from large data sets [2].[2] Neural networks gave solutions to many problems including facial recognition that had stymied vision researchers for decades. The network could associate bitmaps of images with the names of persons contained in them by learning from examples rather than being pre-programmed. Artificial neural networks (ANNs) are the backbone of recent advances in AI. ANNs are arranged into many layers, with outputs of one layer being inputs to the next. Networks with many layers enable what is called "deep learning." The most advanced have hundreds of layers and billions of connections.

Many practical applications use ANNs to classify inputs. For example, a facial recognition ANN is trained on millions of labeled images, effectively grouping them into classes associated with each output label. The ability to classify generalizes to many systems other than facial recognition. Unfortunately, ANNs are prone to fragility—they are likely to give wildly incorrect labels to inputs that were not in their training sets. In addition, ANNs are inscrutable—they are unable to provide an explanation of how an ANN chose a label for an image. The connection structure of the layers is too complex to explain the observed association. Fragility and inscrutability have been major stumbling blocks in generating trust for ANNs in critical applications.

One important reason for the success of ANNs is the hardware on which their network simulations run. Early on researchers realized the transformations made by the layers of the network were mathematically described by the matrix multiplications of linear algebra. It happens that graphics display systems use the same math to manipulate images in real time displays. The graphics industry had developed graphic processing units (GPUs), chips that did the math extremely fast. It was natural to use these same chips to simulate ANNs. The Invidia company, a world leader in graphics chips, has been transformed into the world leader for chips and systems that power ANN models.

Natural language processing. One of the main objectives of AI is to understand natural language by abstracting meaning from text and speech. This has involved statistical analysis of word frequencies and structures in large corpuses of text. Translation between languages is a major challenge; much progress has been made using both logic-based and neuron-based computation. The Turing Test for machine intelligence is tailored for this idea. A machine is considered intelligent if in a natural language conversation, a human observer cannot distinguish it from another human being. To date, there is no consensus among AI researchers that any machine has

[2] In October 2024 the Royal Swedish Academy of Sciences awarded the Nobel Prize in Physics to John J. Hopfield of Princeton University, NJ, USA and Geoffrey E. Hinton of the University of Toronto, Canada "for foundational discoveries and inventions that enable machine learning with artificial neural networks."

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passed the Turing Test. The appearance of ChatGPT in 2022 surprised many by its ability to produce grammatically correct, meaningful and natural sounding text.

Signal processing and computer graphics. Signal processing became important at the end of the nineteenth century for telephony and in the mid twentieth century for high fidelity music and sound reproduction. The underlying mathematical theory, originally developed for analogue electronic devices, is now widely used for sound in digital devices. The mathematics and algorithms for computer graphics developed rapidly in the second half of the twentieth century. These tools play significant roles in automatic generation of images and in the analysis of images for features that can be used for classification.

Programming and software engineering. Almost all AI technologies are combined within systems created by traditional computer programing using the principles of software engineering

Human-computer interaction. Since the 1970s, the computing field has developed a branch of software design to facilitate human-computer interaction (HCI). It seeks to create user interfaces that are easy to utilize and minimize user errors, especially when users are under pressure or duress. HCI was greatly improved by the invention of the mouse in the 1960s, and the commercial touch screens that appeared in the 1990s. Good HCI design incorporates graphics and visualizations that are hard to misinterpret and give good feedback on the progress of work or errors encountered during automated tasks. The well-tested and time-honored principles of HCI are critical to the success of "human machine teaming," a paradigm in which AI augments and amplifies human intelligence rather than replacing it. Good HCI can make a weak program extremely useful, while poor HCI can make a powerful program unusable. Good HCI is essential for AI.

Communications. Many applications of AI involve communication between devices, e.g. using Bluetooth, mobile telephony, and fiber optic technologies. This enables people and organizations to form the networks that are essential for human activities. These include networks of coordination that enable large multinational companies and governments to offer services at large scale. Without reliable communication networks, it would be impossible to manage international trade and supply chains. It has long been realized that the computer has great value in facilitating communications as well as crunching numbers. The principle continues today: AI applications that facilitate communication and coordination are in high demand.

Hardware. With very few exceptions, early expert systems and early neural networks were run on single-processor computers with the so-called von Neumann architecture from 1945. Even with Moore's law doubling the power of these computers about every eighteen months, they were not fast enough for large neural networks. The realization that graphics processing units (GPUs) can perform large chunks of neural network computations provided a special-purpose computer architecture that enabled the construction of Large Language Models. Nvidia soon emerged as the main supplier of chips for generative AI, eclipsing Intel and all its expertise at chips

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for von Neumann architectures. Design engineers are investigating other technologies, such as neuromorphic, memristors, and photonics, to make these models faster and more energy efficient. AI would not have progressed to its impressive state today without the supercomputing enabled by these chips.

The Quest for Artificial General Intelligence

Artificial general intelligence (AGI) refers to machines that can understand or learn any intellectual task that a human being can and perhaps develop all the cognitive abilities of the human brain. But AGI is completely aspirational. There are no known machines that display intelligence or understanding. Machines simply process data, each in its own way. No one knows whether AGI machines can be built, yet many people are pursuing it as their dream.

Julian Togelius' recent book gives a thoughtful analysis of the problems achieving AGI. He spends much of his book analyzing a dozen different interpretations of "artificial" and "intelligence" and concludes that achieving AGI depends on which definition one accepts [4]. For example, "We already have machines with superhuman intelligence, meaning that they are better than most human on some tasks that would seem to require intelligence. Such superhumanly intelligent machines have existed for a long time, decades at least." This includes the payroll systems that, since the 1960s, have computed people's wages taking into account hours worked, overtime payments, tax deductions, pension contributions, and so on. In contrast to these dedicated applications, "you may see a headline that reads 'AI Will Take Over the World,' then look at the AI-powered autocomplete software in your phone and conclude that something so inept is not going to take over the world anytime soon." As Togelius suggests, "It makes you wonder whether the people who write such headlines have ever tried developing an AI system."
In his 2024 book on the "singularity," Ray Kurzweil remains confident that AI machines will pass a very strict and demanding version of the Turing Test by 2029 [5]. He sees that as a prerequisite for AGI. Machines with AGI will be able to adapt to new and previously unseen situations, and to reason and make decisions in complicated and ambiguous situations. Superintelligence is one step beyond AGI with "machine brains that surpass human brains in general intelligence" [6]. Kurzweil is one of the most prominent advocates of the singularity, a point in time around 2045 when machines will become far more intelligent than humans, bringing an end to humanity as we know it. The "end of humanity" he envisions is actually a complete merging of AI technology with human bodies.

This can be compared to the more modest RoboCup challenge to create by 2050 a team of humanoid robots able to beat the human world soccer champion playing to FIFA's rules. Today we are far from meeting even the RoboCup challenge, which appears much easier than achieving AGI and superintelligence.

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Conclusions

Artificial Intelligence is based on a variety of technologies, none of which seek to emulate human intelligence. All the applications developed so far illustrate "narrow AI"—machines that perform specific kinds of human cognitive tasks. A machine that appears intelligent, even superintelligent, for one kind of task often does poorly with other, unrelated kinds of tasks. AGI envisions "broad AI" machines able to perform all human cognitive tasks as well or better than humans. Although AGI has passionate proponents, no one has found a means to approach it. Few expect it to be possible before 2050, if ever.

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